Classical-ML vs. Modern Deblurring under One Protocol: Results for the M1 (data-only) and M2 (physics-aware) Arms

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Abstract

We compare a data-only deep network (M1, NAFNet) against physics-aware classical-ML solvers that know the blur kernel and noise level (M2, led by ADMM with an EPLL Gaussian-mixture prior) on BSDS500 under a single locked protocol: the same images, kernels, noise levels and seeded observations for every method. On the training kernels the two arms are close: NAFNet leads by 0.2–0.5 dB for $\sigma \geq 5/255$, while EPLL leads by 0.9 dB at $\sigma = 1/255$. Under kernel mismatch the picture changes. On unseen motion and Gaussian parameters NAFNet loses 2.9 dB on average and EPLL 0.35 dB, and EPLL wins every noise level, by up to 6.4 dB. A per-noise-level ablation shows that knowing σ is worth at most 0.25 dB to NAFNet, so its deficit comes from not knowing the kernel. We also show that the noiseless ($\sigma = 0$) condition is an *inverse crime* that inflates physics-aware results to 80–100 dB, which disappears once observations are quantized to 8 bits. The unrolled arm (M3) is deferred and not reported here.

1 Setup

Problem. A sharp image x is observed as $y = Ax + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, where A is circular convolution with a point-spread function (PSF) k. Images are BSDS500 luminance in $[0, 1]$; observations are clipped to $[0, 1]$.

Protocol (shared by all methods).

- **Splits.** BSDS500 train (200) / val (100) / test (200). All results below are on the full 200-image test split unless noted.
- **Noise grid.** $\sigma \in \{0, 1, 5, 15, 25\}/255$.
- **PSF sets.** P_1 (training and in-distribution test): Gaussian $\sigma_k \in \{0.7, 1.2, 1.6, 2.0\}$ and linear motion of length $\{9, 15\}$ px at $\{0, 45, 90, 135\}^\circ$ (12 kernels). P_2-interp (unseen parameters, same families): Gaussian $\sigma_k \in \{0.9, 1.4, 1.8\}$ and motion of length $\{11, 13\}$ px at $\{22.5, 67.5, 112.5, 157.5\}^\circ$ (11 kernels). P_2-extrap (unseen family): defocus disks of radius $\{3, 5, 7, 9\}$ px (4 kernels).
- **Observations.** Each (image, PSF, σ) cell has a noise seed derived from its names, so every method sees the identical y and results pair one-to-one: 27 000 test cells per method.
- **Model selection** uses BSDS500 val only. The test split is never used for selection.

M1: data-only. Networks map $y \mapsto \hat{x}$ and receive neither k nor σ. *NAFNet* (width 32, 17.1 M parameters) is the primary model; a plain *U-Net* (7.8 M parameters) is the architecture baseline. Both are trained once on $P_1\times$ all noise levels with 128^2 patches (kernel and noise level drawn uniformly per patch), AdamW, L2 loss, batch 16, learning rate 2×10^{-4} with 1k-iteration warm-up

and cosine decay, 100k iterations. The checkpoint with the highest mean PSNR over the 60-cell grid on a fixed 10-image val subset is kept (NAFNet 29.67 dB at 98k iterations; U-Net 28.43 dB at 94k). Training took 2.8 h and 0.5 h on an RTX 3090 Ti. Five *per-σ specialists* are fine-tuned from the NAFNet checkpoint for 10k iterations each at a single noise level; they measure how much knowing σ would help.

M2: physics-aware classical ML. Solvers receive the true k and σ.

- **Wiener** filter (closed form, noise-level regularisation; not tuned).
- **Richardson–Lucy**, 30 iterations (not tuned).
- **FISTA-TV**: FISTA with an isotropic TV prior, 100 outer and 30 inner ADMM iterations.
- **ADMM-EPLL** (primary): half-quadratic splitting with the EPLL patch prior (Zoran & Weiss, 2011): a 200-component full-covariance GMM on 8×8 zero-mean patches, fitted by EM on 100k patches from BSDS500 train (7 minutes on CPU). Ten iterations; the denoiser strength decreases geometrically from 49/255 to $\max(\sigma, 1/255)$ and the penalty is $\beta_t = \lambda / s_t^2$ (the IRCNN/DPIR schedule). Patch stride 1.

The regularisation weight λ of FISTA-TV and ADMM-EPLL is tuned per noise level on the same 10 val images over P_1: 10 multiples (2^{-5}–2^4) of a default value, chosen by mean val PSNR. M2 therefore uses five scalars per method, against the millions of M1 weights.

Implementation notes.

- Our first ADMM-EPLL used a penalty schedule starting at $\beta_0 = 1$. Since $|\hat{k}(\omega)|^2 \leq 1$, the x-update then barely moved away from the blurred input, and the solver gained under 1 dB even without noise. We replaced it with the schedule above. On the same noiseless motion-blur cell, the corrected solver reaches 35.2 dB instead of 21.9 dB (blurred input: 21.4 dB). Its tuned λ equals the DPIR default $0.23\,\sigma^2$ at every $\sigma > 0$.
- FISTA-TV and ADMM-EPLL were ported operation-for-operation to PyTorch to run on the GPU. Against the NumPy reference on a full-size image, the maximum pixel difference is 10^{-13} (FISTA-TV) and 2×10^{-7} (ADMM-EPLL, float32 patch arithmetic). Per 481×321 image, FISTA-TV takes 2.1 s on the GPU vs. 69 s on one CPU core, and ADMM-EPLL 0.8 s vs. 30–40 s (CPU, stride 2).

Metrics and statistics. PSNR and SSIM against the sharp image, and the data-consistency residual $\mathrm{RMSE}(k * \hat{x} - y)$. Paired comparisons use the *image* as the unit: each image's PSNR is averaged over the PSFs of a set, which gives 200 pairs per (set, σ). We report the mean and median difference, the fraction of images won, and Wilcoxon signed-rank p-values with Holm correction over all 30 tests.

2 Main results

Table 1 and Figure 1 give PSNR for every method, PSF set and noise level $\sigma > 0$. Table 2 gives SSIM. $\sigma = 0$ is treated separately in Section 5.

In distribution (P_1). NAFNet and ADMM-EPLL are within 0.05 dB of each other on average over $\sigma > 0$ (27.48 vs. 27.43 dB), but the ranking depends on the noise level. EPLL wins at $\sigma = 1/255$ (+0.88 dB on every one of the 200 images); NAFNet wins at $\sigma \geq 5/255$ by 0.22–0.46 dB, on 97–100 % of images (Table 3). FISTA-TV is about 1 dB behind NAFNet at $\sigma \geq 5/255$ and tied at $\sigma = 1/255$. The untuned baselines fare poorly once noise is present: Wiener scores below the blurred input at every $\sigma \geq 5/255$, and Richardson–Lucy at $\sigma \geq 15/255$.

Table 1: Mean test PSNR (dB) over 200 images and all PSFs of the set. Best restored result per row in bold. "Blurred" is the unrestored observation.

			M2 (knows k, σ)				M1 (data only)	
Set	σ	Blurred	Wiener	R–L	FISTA-TV	ADMM-EPLL	U-Net	NAFNet
P_1	1	24.52	25.38	27.98	31.00	**31.89**	29.56	31.00
	5	23.93	23.37	25.10	27.02	27.82	27.44	**28.04**
	15	21.26	21.09	19.27	24.91	25.51	25.56	**25.92**
	25	18.69	18.38	15.80	23.98	24.49	24.64	**24.95**
P_2-interp	1	24.09	24.74	27.27	30.34	**31.24**	24.92	24.85
	5	23.58	23.53	25.11	26.76	**27.48**	24.72	24.79
	15	21.09	21.16	19.56	24.75	**25.29**	24.35	24.51
	25	18.60	18.41	16.04	23.82	**24.32**	23.94	24.13
P_2-extrap	1	22.57	23.27	24.74	27.67	**28.83**	23.00	22.90
	5	22.22	22.97	23.98	24.86	**25.50**	23.58	23.62
	15	20.26	20.74	21.05	23.48	**23.91**	23.35	23.44
	25	18.08	18.06	18.51	22.79	**23.27**	23.02	23.15

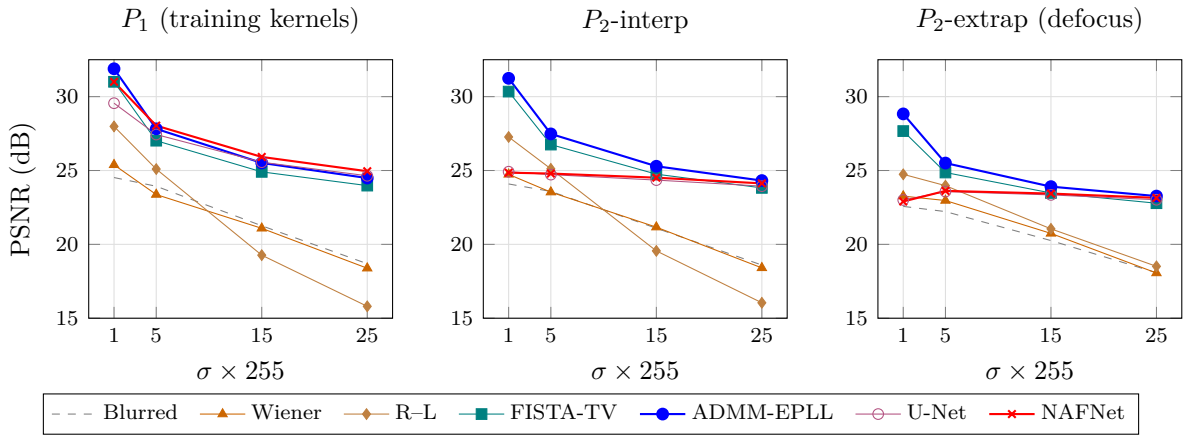


Figure 1: Mean test PSNR vs. noise level. M2 in blue/teal, M1 in red/magenta. On P_1 NAFNet and ADMM-EPLL nearly coincide; on both P_2 sets the M1 curves collapse towards the blurred input at low noise, while the M2 solvers keep their gain.

Under kernel mismatch (P_2). ADMM-EPLL is best in every row of both P_2 sets. The margin over NAFNet is largest at low noise (+6.4 dB on P_2-interp and +5.9 dB on P_2-extrap at $\sigma = 1/255$) and shrinks to +0.1–0.2 dB at $\sigma = 25/255$, where noise rather than blur limits everyone. Every EPLL-vs-NAFNet difference on P_2 is significant after Holm correction ($p < 10^{-4}$). Even TV, a prior with essentially one parameter, beats both networks on P_2 up to $\sigma = 15/255$.

Table 2: Mean test SSIM. Best restored result per row in bold.

Set	σ	Wiener	R–L	FISTA-TV	ADMM-EPLL	U-Net	NAFNet
P_1	1	0.584	0.808	0.878	**0.900**	0.861	0.889
	5	0.492	0.599	0.754	0.790	0.785	**0.802**
	15	0.460	0.285	0.660	0.692	0.702	**0.717**
	25	0.428	0.171	0.618	0.643	0.657	**0.672**
P_2-interp	1	0.556	0.795	0.871	**0.894**	0.694	0.691
	5	0.498	0.603	0.748	**0.780**	0.686	0.690
	15	0.464	0.288	0.655	**0.682**	0.656	0.666
	25	0.431	0.172	0.611	0.635	0.628	**0.640**
P_2-extrap	1	0.475	0.672	0.777	**0.820**	0.593	0.589
	5	0.480	0.580	0.656	**0.687**	0.612	0.616
	15	0.493	0.347	0.590	**0.613**	0.593	0.600
	25	0.482	0.226	0.558	0.582	0.574	**0.583**

Table 3: Paired comparison of each M2 solver against NAFNet (difference = M2 − NAFNet, dB). Unit: image (PSNR averaged over the set's PSFs), $n = 200$. "Win" is the percentage of images where the M2 solver is better; p is the Wilcoxon signed-rank test, Holm-adjusted over 30 tests (the $\sigma = 0$ tests are included in the correction but not shown).

		ADMM-EPLL − NAFNet				FISTA-TV − NAFNet			
Set	σ	mean	median	win %	p_{Holm}	mean	median	win %	p_{Holm}
P_1	1	+0.88	+0.82	100	$< 10^{-4}$	+0.00	+0.05	57	0.279
	5	-0.22	-0.19	3	$< 10^{-4}$	-1.02	-0.93	0	$< 10^{-4}$
	15	-0.40	-0.36	0	$< 10^{-4}$	-1.01	-0.93	0	$< 10^{-4}$
	25	-0.46	-0.42	0	$< 10^{-4}$	-0.97	-0.91	0	$< 10^{-4}$
P_2-interp	1	+6.39	+6.34	100	$< 10^{-4}$	+5.49	+5.43	100	$< 10^{-4}$
	5	+2.68	+2.66	100	$< 10^{-4}$	+1.96	+1.96	100	$< 10^{-4}$
	15	+0.77	+0.75	100	$< 10^{-4}$	+0.24	+0.29	78	$< 10^{-4}$
	25	+0.19	+0.18	82	$< 10^{-4}$	-0.30	-0.25	2	$< 10^{-4}$
P_2-extrap	1	+5.93	+5.79	100	$< 10^{-4}$	+4.77	+4.69	100	$< 10^{-4}$
	5	+1.88	+1.82	100	$< 10^{-4}$	+1.24	+1.17	100	$< 10^{-4}$
	15	+0.47	+0.45	99	$< 10^{-4}$	+0.04	+0.06	64	0.0003
	25	+0.12	+0.13	83	$< 10^{-4}$	-0.36	-0.33	0	$< 10^{-4}$

3 Kernel mismatch

Table 4 summarises each method's PSNR averaged over $\sigma > 0$ on each set, the change from P_1, and the gain over the blurred input. The gain column corrects for the sets being of different difficulty: the blurred input itself is 0.26 dB worse on P_2-interp and 1.32 dB worse on P_2-extrap.

The M2 solvers keep almost all of their gain on unseen kernels: ADMM-EPLL goes from +5.33 dB on P_1 to +5.24 dB on P_2-interp and +4.60 dB on the unseen defocus family. NAFNet's gain halves, from +5.38 to +2.73 and +2.49 dB, and U-Net behaves the same way. The two architectures differ by about 0.7 dB on P_1 but by less than 0.1 dB on P_2: under mismatch, the better architecture does not help.

Where M1 fails. Table 5 splits the results by kernel family. On Gaussian kernels NAFNet generalises well: unseen widths cost it 0.7 dB, and it stays within 0.2 dB of EPLL. The failure is on motion blur. Motion kernels at the unseen angles $22.5°, 67.5°, \ldots$ cost NAFNet 3.6 dB, leaving it essentially at the blurred input's level (23.40 vs. 21.28 dB, while EPLL reaches 26.78 dB). The network learned to undo the eight training motion directions, not motion blur in general. On

Table 4: Kernel-mismatch summary, mean over $\sigma \in \{1, 5, 15, 25\}/255$. Δ is the change from P_1; "gain" is PSNR minus the blurred input.

		P_1	P_2-interp		P_2-extrap		Gain over blurred		
Method	Arm	PSNR	PSNR	Δ	PSNR	Δ	P_1	P_2-int	P_2-ext
Blurred input	–	22.10	21.84	-0.26	20.78	-1.32	–	–	–
Wiener	M2	22.06	21.96	-0.10	21.26	-0.80	-0.04	+0.12	+0.48
Richardson–Lucy	M2	22.04	22.00	-0.04	22.07	+0.03	-0.06	+0.16	+1.29
FISTA-TV	M2	26.73	26.42	-0.31	24.70	-2.03	+4.63	+4.58	+3.92
ADMM-EPLL	M2	27.43	27.08	-0.35	25.38	-2.05	+5.33	+5.24	+4.60
U-Net	M1	26.80	24.48	-2.32	23.24	-3.56	+4.70	+2.64	+2.45
NAFNet	M1	27.48	24.57	-2.91	23.28	-4.20	+5.38	+2.73	+2.49

defocus, which it never saw, NAFNet recovers 2.5 dB, against EPLL's 4.6 dB.

Table 5: Mean test PSNR by kernel family (mean over $\sigma > 0$). Best in bold.

Set	Family	Blurred	Wiener	R–L	FISTA-TV	ADMM-EPLL	U-Net	NAFNet
P_1	gaussian	23.52	23.20	23.43	27.45	28.14	27.94	**28.43**
P_1	motion	21.39	21.48	21.34	26.37	**27.07**	26.23	27.00
P_2-interp	gaussian	23.34	23.01	23.41	27.29	**27.89**	27.34	27.70
P_2-interp	motion	21.28	21.57	21.46	26.09	**26.78**	23.41	23.40
P_2-extrap	defocus	20.78	21.26	22.07	24.70	**25.38**	23.24	23.28

Data consistency. Table 6 reports RMSE$(k * \hat{x} - y)$ in units of 1/255. A restoration consistent with the observation should leave a residual close to the noise level σ. On P_1, FISTA-TV, ADMM-EPLL and both networks are close to it. Under mismatch at low noise, the M1 outputs are inconsistent with the observation: at $\sigma = 1/255$ on P_2-interp, NAFNet's residual is 3.9, against 0.9 for FISTA-TV and ADMM-EPLL. In other words, the network returns an image that, re-blurred with the true kernel, does not reproduce what was observed. The M2 solvers enforce this consistency by construction.

Table 6: Data-consistency residual RMSE$(k * \hat{x} - y) \times 255$. For a consistent estimate this is about $\sigma \times 255$.

Set	σ	Wiener	R–L	FISTA-TV	ADMM-EPLL	U-Net	NAFNet
P_1	1	0.93	1.21	0.87	0.85	1.52	1.13
	5	4.54	4.23	4.60	4.74	4.91	4.78
	15	18.00	12.42	14.27	14.54	14.51	14.46
	25	32.65	20.66	24.13	24.08	23.98	23.97
P_2-interp	1	1.05	1.26	0.90	0.88	4.13	3.92
	5	4.70	4.40	4.68	4.80	6.34	6.10
	15	18.27	12.87	14.38	14.62	15.01	14.94
	25	32.95	21.26	24.19	24.16	24.30	24.28
P_2-extrap	1	1.24	1.37	0.99	0.94	2.89	2.95
	5	5.20	4.84	4.92	4.91	5.58	5.46
	15	19.46	14.13	14.71	14.73	14.95	14.90
	25	34.40	23.25	24.39	24.28	24.41	24.38

4 Ablations

Does M1 lose because it does not know σ? M2 is given the true noise level; M1 is not. Table 7 compares the single NAFNet with specialists fine-tuned for one noise level each and evaluated at that level. Knowing σ gains at most 0.25 dB (on P_1 at $\sigma = 1/255$), essentially nothing elsewhere, and slightly hurts on P_2. The noise level is not the missing information. The kernel is, which only physics-aware methods use.

Table 7: Per-noise-level oracle: single NAFNet vs. a specialist fine-tuned at that σ (test PSNR, dB).

	P_1			P_2-interp			P_2-extrap		
σ	single	spec.	Δ	single	spec.	Δ	single	spec.	Δ
1	31.00	31.25	+0.25	24.85	24.75	-0.11	22.90	22.80	-0.10
5	28.04	28.11	+0.08	24.79	24.72	-0.07	23.62	23.58	-0.04
15	25.92	25.94	+0.02	24.51	24.51	-0.01	23.44	23.40	-0.03
25	24.95	24.97	+0.02	24.13	24.12	-0.01	23.15	23.16	+0.01

Prior capacity (GMM K-sweep). Table 8 and Figure 2 run ADMM-EPLL with GMM priors of $K \in \{1, 4, 16, 64, 200\}$ components on a seeded 50-image subset of the test split, all with the λ values tuned for $K = 200$. Capacity helps quickly and then saturates. Going from a single Gaussian ($K = 1$, a linear patch prior) to $K = 4$ components adds 1.8 dB on P_1 (mean over $\sigma > 0$); further steps add 0.27 dB ($4 \to 16$), 0.11 dB ($16 \to 64$) and 0.02 dB ($64 \to 200$). The same curve appears on both P_2 sets, and the gap between P_1 and P_2-extrap stays between 1.6 and 2.1 dB for every K: a deeper GMM prior improves the M2 solver uniformly but does not change its robustness to kernel mismatch, which comes from the known forward operator. Within the GMM family, $K = 200$ is past the point of diminishing returns.

Table 8: ADMM-EPLL with K-component GMM priors (50-image test subset, test PSNR in dB). Left: P_1 per noise level. Right: mean over $\sigma > 0$ per set.

	P_1, σ				Mean over $\sigma > 0$		
K	1	5	15	25	P_1	P_2-int	P_2-ext
1	28.92	25.44	23.80	23.15	25.33	25.03	23.69
4	31.35	27.49	25.33	24.37	27.13	26.82	25.19
16	31.74	27.78	25.54	24.53	27.40	27.07	25.40
64	31.92	27.90	25.62	24.60	27.51	27.18	25.47
200	31.96	27.92	25.63	24.60	27.53	27.19	25.48

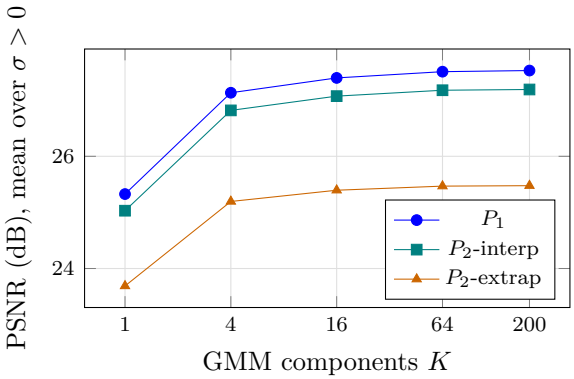


Figure 2: ADMM-EPLL test PSNR as a function of prior capacity.

5 The noiseless case is an inverse crime

At $\sigma = 0$ the test observations are produced with exactly the circular FFT operator that the M2 solvers invert, with no noise and no quantization. Wiener deconvolution then inverts the blur almost exactly, scoring 80–100 dB, and the other M2 solvers are inflated as well. Such numbers say nothing about real deblurring. As a check we rounded the $\sigma = 0$ observations to 8-bit levels, an error of about 0.0011 RMS (Table 9). Wiener collapses to 5–13 dB, far below the blurred input, because it amplifies the quantization error without limit. ADMM-EPLL, whose $\sigma = 0$ weight was tuned to almost no regularisation, loses 6–9 dB. The networks and Richardson–Lucy are essentially unaffected. Even on quantized data, though, ADMM-EPLL and FISTA-TV stay well ahead of NAFNet on P_2 (e.g. 32.0 and 32.5 vs. 24.8 dB on P_2-interp).

We therefore exclude $\sigma = 0$ from every average in this report and do not use it for any M1-vs-M2 claim. In future, observations should be quantized, or a noise floor added, as part of the protocol.

Table 9: $\sigma = 0$: test PSNR (dB) on the exact observations vs. the same observations rounded to 8 bits. The λ values tuned for $\sigma = 0$ are kept.

	P_1		P_2-interp		P_2-extrap	
Method	exact	8-bit	exact	8-bit	exact	8-bit
Blurred input	24.55	24.55	24.11	24.11	22.59	22.59
Wiener	100.49	12.59	82.70	6.66	80.31	5.41
Richardson–Lucy	28.44	28.38	27.43	27.41	24.78	24.78
FISTA-TV	34.22	33.52	33.03	32.51	29.96	29.62
ADMM-EPLL	41.00	33.58	38.89	32.04	36.09	27.30
U-Net	30.29	30.17	25.05	25.01	22.70	22.73
NAFNet	32.61	32.09	24.90	24.79	22.72	22.71

6 Discussion

1. **In distribution, a 17 M-parameter network and a 200-component GMM with five tuned scalars are on par.** NAFNet's advantage on P_1 is at most 0.5 dB and appears only at moderate-to-high noise. The learned prior (EPLL) is worth about 0.7 dB over the hand-crafted one (TV).

2. **Physics-awareness is what transfers.** Under kernel mismatch the M1 networks lose half their gain, and on unseen motion directions nearly all of it, while the M2 solvers barely degrade. The per-σ oracle and the residual analysis both point to the unknown kernel, not the unknown noise level, as the cause.

3. **Architecture matters only in distribution.** NAFNet beats U-Net by 0.7 dB on P_1 and by less than 0.1 dB on P_2.

4. **The gap closes with noise.** At $\sigma = 25/255$ all good methods lie within about 1 dB on every set: the problem is noise-limited, and knowing the kernel helps less.

Fairness and limitations.

- M2 is given the true k and σ (non-blind), and its λ is tuned per noise level on val. This is the intended contrast between the arms, not a confound. A blind or estimated-kernel M2 is not evaluated.
- M3 (unrolled, physics-aware networks) is not yet trained. It is needed to separate physics-awareness from prior depth.
- The noise model is synthetic Gaussian with circular boundaries; real blur (GoPro) is not evaluated.
- Each model was trained once (one seed); differences of a few tenths of a dB between M1 variants should be read with that in mind.
- LPIPS was not computed for all methods and is omitted.
- The K-sweep reuses the λ tuned for $K = 200$ and runs on 50 images. For ADMM-EPLL at $\sigma = 0$, the best λ lay at the edge of the search grid (an inverse-crime effect, Section 5).

Reproducibility. Training: `scripts/train_m1.py`; GMM: `scripts/train_gmm.py`; evaluation: `scripts/evaluate.py` (M2 on the GPU with `--m2-backend torch`); tables and statistics in this report: `scripts/analyze_m1_m2.py`. Protocol manifests (PSF sets, splits, val subset) are in `artifacts/protocol/`.